ECE2191 Cheat Sheet

Probability and AI for Engineers. Methods, conditions, traps and past-paper answers. Use with the formula sheet and Z-table. From your weekly notes, practice papers 1–6, the mid-sem, the 2026 formula sheet and the Z-table. Worked numbers rechecked in Python.

1 How to score, and what is handed to you

Read once before you start.

Every answer: formula → substitution → number → one sentence of meaning

Trap: "No marks if no reasoning." Text boxes: a/b, a^(b), C(n,r), sqrt(), Phi(x).

Formula sheet gives: norms, inner product, $W = (X^TX)^{-1}X^TY$, MSE, R^2, BIC, GD update, sigmoid, CE, softmax, activations, losses, quadratic formula, derivative rules

Trap: It gives NO probability formulas. Sections 2 to 8 below are what you must bring.

Not on the formula sheet: $\sigma' = \sigma(1-\sigma)$, ridge, precision/recall/F1, X^+ for fat X, C(n+k,k)

Z-table gives $\Phi(z) = P(Z \le z)$, 4 decimals, row = 1st decimal, column = 2nd

Needs: Q(z) = P(Z>z) = 1−Φ(z). Φ(−z) = 1−Φ(z).

Trap: Older papers gave a Q-table. The 2026 table is Φ.

Φ(1)=0.8413 Φ(1.5)=0.9332 Φ(1.6)=0.9452 Φ(1.96)=0.9750 Φ(2)=0.9772 Φ(2.576)≈0.995 Φ(0.84)=0.7995

z for a two-sided CI: 90% 1.645, 95% 1.96, 99% 2.576

2 Which tool? Read the question, pick the section

Seats, committees, "exactly k faulty in a batch" → 3. Event trees, "given that" → 3

Fixed n trials → Binomial. Until first success → Geometric. Count per time → Poisson. Wait time → Exponential (4)

PDF with unknown constant, CDF, E, Var, Y=aX+b → 5

Joint table or joint PDF, independent?, uncorrelated?, conditional → 6

Gaussian "probability exceeds", sample mean, CLT → 7. CI, sample size → 8. Decide 0/1 from noisy y → 9

Fit, residuals, R^2, BIC, features → 10. Learning rate, iterations → 11. Classes, metrics, loss → 12. ReLU nets, ridge → 13

3 Probability rules and counting

practice 2 Q6, 3 Q3, 4 Q1, Q6

$P(A|B) = P(A \cap B)/P(B)$. $P(A \cap B) = P(A|B)P(B)$

Trap: $P(A|B) \neq P(B|A)$.

Total: $P(A) = \Sigma\, P(A|B_i)P(B_i)$. Bayes: $P(B_i|A) = P(A|B_i)P(B_i)/P(A)$

Needs: B_i partition the space.

Trap: Draw the tree. Leaves sum to 1.

Independent $\Leftrightarrow$ $P(A \cap B) = P(A)P(B)$

C(n,k) = n!/(k!(n−k)!). Arrangements in a line n!. Round table (n−1)!

Hypergeometric: P(j special in k drawn from N with M special) = C(M,j)C(N−M,k−j)/C(N,k)

Needs: No replacement.

Groups by category: Π C(category size, chosen) / C(total, group size)

Trap: Chosen counts must add to the group size.

Past paper: Spouse next to me at a 20-seat round table: 2·18!/19! = 2/19. Exactly 1 of 5 chips faulty (11 of 30 broken): C(11,1)C(19,4)/C(30,5) = 0.2992.

Past paper: P(not faulty) = 0.8(5/30)+0.6(25/30) = 0.6333; P(new | not faulty) = 0.1333/0.6333 = 0.2105.

4 Discrete distributions

practice 2 Q1, 3 Q2, 4 Q4, mid-sem Q1–2

Model	Use when	P(X=k) or f(x)	E	Var
Bernoulli(p)	one yes/no trial	p, 1−p	p	p(1−p)
Binomial(n,p)	fixed n, independent, same p	$C(n,k)p^k(1-p)^{n-k}$	np	np(1−p)
Geometric(p)	trials until 1st success	$(1-p)^{k-1}p$, k≥1	1/p	(1−p)/p²
Poisson(λ)	count per interval, rate given	$e^{-\lambda}\lambda^k/k!$	λ	λ
Hypergeom.	draw k without replacement	C(M,j)C(N−M,k−j)/C(N,k)	kM/N	—
Uniform(a,b)	equally likely in a range	1/(b−a)	(a+b)/2	(b−a)²/12
Exponential(λ)	waiting time, constant rate	$\lambda e^{-\lambda x}$	1/λ	1/λ²
Gaussian(μ,σ²)	noise, many small effects	$e^{-(x-\mu)^2/2\sigma^2}/(\sigma\sqrt{2\pi})$	μ	σ²

Memoryless: only Geometric (discrete) and Exponential (continuous)

Binomial(n,p): $P(k) = C(n,k)p^k(1-p)^{n-k}$, E = np, Var = np(1−p)

Needs: Fixed n, independent, same p.

Trap: P(at least one) = $1-(1-p)^n$. Bursty errors ⇒ not binomial, but E[S] = np still holds.

Geometric(p): $P(k) = (1-p)^{k-1}p$, $k \ge 1$. $P(X>k) = (1-p)^k$. E = 1/p, Var = $(1-p)/p^2$

Needs: Trials up to and including first success.

Trap: Memoryless: 5 failures done, P(more than 3 more) = 0.8^3.

Poisson(λ): $P(k) = e^{-\lambda}\lambda^k/k!$, E = Var = λ

Needs: λ = rate × interval (same units).

Trap: Scale λ first: 600/min over 200 ms ⇒ λ = 2. 1460/yr over a day ⇒ λ = 4.

$P(N=0) = e^{-\lambda}$. $P(N \le 2) = e^{-\lambda}(1+\lambda+\lambda^2/2)$

Trap: λ=2: 0.1353, 0.6767. λ=4: 0.0183, 0.2381.

$E[X] = \Sigma x\, p(x)$. $Var = E[X^2] - (E[X])^2$

Needs: PMF sums to 1.

Y = g(X) discrete: P(Y=y) = sum of P over the x values (or regions) that map to y

Past paper: Frisbee (p=0.1): $P(X=k) = 0.9^{k-1}(0.1)$; $P(X>3) = 0.9^3 = 0.729$; E=10. Meatballs in 100 throws: Binomial(100, 0.1).

5 Continuous: PDF, CDF, unknown constant

practice 2 Q3, 3 Q4, 4 Q2, mid-sem Q4–5, Q13–14

$\int f = 1$ fixes the constant. A second fact (e.g. E[X]) fixes a second constant

F(x) = ∫ from the lower end to x of f. Write 0 below and 1 above

Needs: Piecewise answer.

Trap: P(X = x) = 0 for continuous X.

P(a<X<b) = F(b)−F(a). $E[X] = \int xf$. $E[X^2] = \int x^2 f$

Trap: Symmetric f ⇒ E = centre, P(X ≥ centre) = ½.

Y = aX+b: E = aE[X]+b, $Var = a^2Var(X)$

Trap: °F = 1.8°C+32 ⇒ $Var_F = 3.24\, Var_C$. Y = 2X+1 ⇒ Var ×4.

Exponential(λ): $f = \lambda e^{-\lambda x}$, $F = 1-e^{-\lambda x}$, $P(X>x) = e^{-\lambda x}$, E = 1/λ, $Var = 1/\lambda^2$

Needs: Constant failure rate.

Trap: Mean 200 h ⇒ λ = 0.005, Var = 40,000, SD = 200. "Most fail near 200 h" is FALSE: density is largest at 0.

Gaussian N(μ,σ²): $f = e^{-(x-\mu)^2/2\sigma^2}/(\sigma\sqrt{2\pi})$, peak $1/(\sigma\sqrt{2\pi})$

Trap: Higher peak ⇒ smaller σ. f(0) is not a probability.

Flat CDF on an interval ⇒ zero probability there

Trap: Not uniform.

Past paper: f = a(4−(x−38)²) on [36,40] ⇒ a = 3/32, E = 38, P(X ≥ 38) = ½. f = ax^b on [0,1], E=0.6 ⇒ a=1.5, b=0.5, F = $x^{1.5}$, Var = 3/7−0.36 = 0.0686, P(X>0.5) = 0.646.

6 Joint distributions, covariance, vectors

practice 3 Q5, 4 Q5–6, mid-sem Q3, Q15

Marginal: $p_X(x) = \Sigma_y\, p(x,y)$, $f_X(x) = \int f(x,y)dy$

Trap: Limits for y may depend on x.

Conditional: $f(y|x) = f(x,y)/f_X(x)$

Trap: f=2 on 0<y<x<1 ⇒ f_X=2x, f(y|x)=1/x, P(Y<x/2 | X=x) = ½.

Claim	Test	Implies
Independent	$p(x,y) = p_X\, p_Y$ for EVERY cell (support a rectangle)	uncorrelated
Uncorrelated	Cov = E[XY] − E[X]E[Y] = 0	NOT independence (unless jointly Gaussian)
Orthogonal	E[XY] = 0	uncorrelated only if a mean is 0

$Cov = E[XY] - E[X]E[Y]$. $\rho = Cov/(\sigma_X\sigma_Y) \in [-1,1]$

Trap: Cov = 0 ⇒ uncorrelated, NOT independent (X on {−1,0,1}, Y = X²).

E[aX+bY] = aE[X]+bE[Y] (always). Var(X+Y) = VarX + VarY + 2Cov

Trap: Cov(F, aF+b) = a Var(F).

Vector: $C = E[(X-\mu)(X-\mu)^T] = R - \mu\mu^T$, symmetric PSD

Needs: Diagonal = variances.

Jointly Gaussian and Cov = 0 ⇒ independent

Needs: Both needed.

Past paper: p(0,1)=p(1,0)=p(2,1)=1/3: E[X]=1, E[Y]=2/3, E[XY]=2/3 ⇒ Cov = 0, uncorrelated. P(0,0)=0 ≠ (1/3)(1/3) ⇒ dependent.

Past paper: 3 balls from 2R,3W,4B: p(x,y) = C(2,x)C(3,y)C(4,3−x−y)/C(9,3).

7 Gaussians, sample means, LLN, CLT

practice 2 Q2, 3 Q1, Q6, 4 Q7, mid-sem Q6–8

$Z = (X-\mu)/\sigma$. $P(X \le x) = \Phi(z)$. $P(X > x) = 1-\Phi(z)$

Trap: Divide by σ, not σ².

$\bar{X} \sim N(\mu, \sigma^2/n)$: use $SE = \sigma/\sqrt{n}$. Sum $S_n \sim N(n\mu, n\sigma^2)$

Needs: Gaussian data, or CLT for large n.

Trap: A total uses the sum form.

CLT: IID, finite mean and variance; n ≥ 30 is a rule of thumb

Trap: Rare events (p = 0.001, n=30): $P(\bar{X}=0) = 0.97$ ⇒ not Gaussian.

LLN: $\bar{X} \to \mu$ (where). CLT: shape around μ

Needs: IID needed.

Trap: Drift or bursts break IID: more data does not fix it.

Symmetric interval with prob p: $\mu \pm z\cdot\sigma/\sqrt{n}$ with $\Phi(z) = (1+p)/2$
Trap: 95% ⇒ 1.96. 60% ⇒ $\Phi(z)=0.8$ ⇒ z = 0.84.

Past paper: Sources above 240: A (230,5) z=2 ⇒ 0.0228; B (225,10) z=1.5 ⇒ 0.0668 ⇒ B more likely. Sensor lifetimes ≥30,000: S1 0.0228, S2 0.0668 ⇒ choose S2.

Past paper: Mid-sem Q8: (13.2−12)/(6/√64)=1.6 ⇒ 1−0.9452=0.0548. Chef: σ=10, n=16 ⇒ SE 2.5; P(45<$\bar{X}$<52)=Φ(2.8)−0.5=0.4974; 95% of means in 45±4.9.

8 Estimation, confidence intervals, sample size

practice 2 Q4–5, mid-sem Q9–12

$\bar{x}$ estimates μ. $s^2 = \Sigma(x_i-\bar{x})^2/(n-1)$
Trap: Divide by n−1.

Unbiased: $E[\hat{\theta}] = \theta$. Consistent: $\hat{\theta} \to \theta$ as n grows
Trap: X_1 and $\bar{X}$ are both unbiased; $\bar{X}$ has the smaller SE.

Situation	Use	Conditions
P(one reading > t)	$1-\Phi((t-\mu)/\sigma)$	X Gaussian
P(sample mean > t)	$1-\Phi((t-\mu)/(\sigma/\sqrt{n}))$	Gaussian data or large n (CLT)
P(total > t)	$1-\Phi((t-n\mu)/(\sigma\sqrt{n}))$	same
CI for μ	$\bar{x} \pm z\cdot\sigma/\sqrt{n}$	IID, σ known
σ unknown	$\bar{x} \pm t\cdot s/\sqrt{n}$ (Student t)	z is not exact
Sample size	$n = (z\sigma/E)^2$, round up	σ known

Meaning: 95% of intervals built this way contain μ
Trap: Not "95% of items" and not "μ has 95% probability".

CI fully inside a tolerance ⇒ supports "mean is within tolerance"

Higher confidence ⇒ wider. Larger n ⇒ narrower

$n = (z\,\sigma\,/\,E)^2$, ALWAYS round up
Needs: E = maximum margin of error.
Trap: Halve E ⇒ 4× n.

Past paper: $P(|\bar{X}-\mu| < \sigma/10) \geq 0.95 \Rightarrow 1.96\sigma/\sqrt{n} \leq \sigma/10 \Rightarrow n \geq 384.16 \Rightarrow n = 385$.

9 Binary detection in Gaussian noise

practice 4 Q3, practice 5

Decide "0" (X=−1) if $f(y|-1)P_0 > f(y|+1)P_1$
Needs: Y = X + N, N ~ N(0,σ²).

Threshold $T = (\sigma^2/2)\ln(P_0/P_1)$: decide 0 if y < T
Needs: Equal noise, levels ±1.
Trap: Equal priors ⇒ T = 0.

$P(\text{error}) = P_0\cdot Q((T+1)/\sigma) + P_1\cdot Q((1-T)/\sigma)$
Trap: Q = 1−Φ.

Past paper: $P_0 = 2P_1$, σ² = 1/16: T = ln2/32 = 0.0217; P(error) ≈ 3.0×10^{-5}.

10 Regression

practice 5, 6 Q1–11

Univariate: $w = \hat{\sigma_{xy}}/\hat{\sigma_{xx}}$, $b = \bar{y} - w\,\bar{x}$ (or $w = (\Sigma xy - m\,\bar{x}\bar{y})/(\Sigma x^2 - m\,\bar{x}^2)$)
Needs: Divide by m in $\hat{\sigma}$.

Least squares with bias ⇒ $\Sigma e_i = 0$, $\Sigma x_i e_i = 0$, $\Sigma \hat{y}_i e_i = 0$
Needs: $e_i = y_i - \hat{y}_i$.
Trap: Use these to recover lost residuals.

$MSE = (1/m)\Sigma e_i^2$. $R^2 = 1 - SSE/SST$ (fraction of variance explained)
Trap: Higher R^2 is better on training data, but always grows with more features.

Tall X (m ≥ n): $W = (X^TX)^{-1}X^TY$. Fat X: $w = X^T(XX^T)^{-1}y$ (minimum norm)
Needs: Add a column of 1s for the bias.
Trap: Fat: infinitely many zero-error fits; zero training error means little.

Linear in parameters ⇒ still linear regression (x^2, cos x, sin x features)

Polynomial features of degree ≤ k from n inputs: C(n+k, k)
Trap: n=3: k=2 ⇒ 10, k=3 ⇒ 20. 20 params with 12 samples ⇒ underdetermined.

BIC = d ln m + m ln MSE: lower is better
Trap: Can be negative. Only the minimum matters.

Feature rescaling: LS predictions unchanged; ridge changes
Trap: x′=2x−1, y′=10y+5 ⇒ w′=5w, R^2 same, MSE ×100.

Fit y on x and x on y: $w_1 w_2 = R^2$
Trap: Lines coincide only if $R^2 = 1$.

Outliers pull the line; check residuals, remove bad data or refit

Past paper: BIC, m=50: A (d=3, MSE 0.40) = −34.08, B (d=6, 0.33) = −31.96 ⇒ A. B wins only if MSE < 0.316. With m=5000 ⇒ B.

Past paper: Data x=1..5, y=2,3,5,4,6: $\hat{y} = 1.3 + 0.9x$; $w_1 w_2 = 0.81 = R^2$.

11 Gradient descent

practice 5, 6 Q12–16

$w \leftarrow w - \eta\nabla L$; stop when $\|\nabla L\|$ small, loss change small, or max iterations
Needs: η > 0.

Quadratic $f = (\lambda/2)(w-w^*)^2 \Rightarrow w^{(t)}-w^* = (1-\eta\lambda)^t(w^{(0)}-w^*)$
Trap: Converges iff $|1-\eta\lambda| < 1$, i.e. $0 < \eta < 2/\lambda$. Best one-step $\eta = 1/\lambda$.

Several curvatures: $\eta < 2/\lambda_{max}$
Trap: $\frac{1}{2}(w_1^2+9w_2^2) \Rightarrow 0 < \eta < 2/9$.

η too small: slow. Too big: oscillates or diverges. Plateau: decay η
Trap: Faster smooth drop after a change ⇒ η was increased.

$\sigma' = \sigma(1-\sigma)$. Chain rule: $dy/dw_1 = (dy/d\theta_2)(d\theta_2/d\theta_1)(d\theta_1/dw_1)$

Past paper: $f=(w-2)^2+1$, $w_0=5$, $w_1=3.8 \Rightarrow \eta=0.2$; $w_2=3.08$, $w_3=2.648$; $|w-2| < 0.01$ needs t=12. η=1 oscillates 5, −1.

Past paper: $f=\frac{1}{2}x^2-4x+12$ from $x_0=-2 \Rightarrow \eta=1$ lands on x=4. $\nabla[(x_1-3)^3\sigma(x_2)]$ at (−1,1) = (35.09, −12.58).

12 Classification

practice 5, 6 Q17–22

Logistic: $\hat{y} = \sigma(w^Tx)$, class 1 if $\hat{y} \geq 0.5 \Leftrightarrow w^Tx \geq 0$
Needs: Boundary is a line or hyperplane.

BCE: $-y\ln\hat{y} - (1-y)\ln(1-\hat{y})$. Gradient $(\hat{y}-y)x$
Needs: Convex.
Trap: Largest $|\hat{y}-y| \Rightarrow$ largest update. MSE + sigmoid stalls.

Softmax $p_r = e^{z_r}/\Sigma e^{z_i}$; CCE = −ln p_true
Needs: One-hot labels.
Trap: $z_i/\Sigma z$ fails with negatives. Integer labels imply false order.

Only logistic allowed, K classes ⇒ K one-vs-rest models
Trap: 3 classes ⇒ 3 models.

Boundary $w_1x_1+w_2x_2+b = 0$; w points to class +1
Trap: Plug a point (origin) to find the side.

Metric	Formula	Choose it when
Accuracy	(TP+TN)/N	classes balanced
Precision	TP/(TP+FP)	false alarms are costly
Recall	TP/(TP+FN)	misses are dangerous (lower the threshold)
F_1	2PR/(P+R)	imbalanced classes

Hinge max(0,1−yf): y=−1 minimised for $f \leq -1$. Log loss $\ln(1+e^{-yf})$: no finite minimiser

Past paper: Softmax (5.2, 3.6, −2.7) ⇒ (0.832, 0.168, 0.0003). TP=6, FP=2, FN=3 ⇒ P=0.75, R=0.667, F_1=0.706.

13 Neural nets and ridge

practice 6 Q23+

Neuron $\varphi(w^Tx)$; with σ it is logistic regression

$h = \varphi(W_1^Tx)$, $y = W_2^Th$. Params ≈ nH + Hp (+ biases)
Trap: 100·500 + 500·1 = 50,500.

ReLU [0,∞), sigmoid (0,1), tanh (−1,1)

No activations ⇒ the whole MLP collapses to one linear map

ReLU(x) − 2ReLU(x−1) + ReLU(x−2): triangle, peak 1 at x=1, zero outside [0,2]
Needs: One kink per hidden unit.

Ridge: $L = MSE + \alpha\|w\|^2$, $w = (X^TX + \alpha I)^{-1}X^Ty$
Needs: α ≥ 0.
Trap: Larger α shrinks w. Sensitive to feature units.

14 Concept traps (mid-sem multiple choice)

mid-sem

Linearity of expectation needs no independence; variance additivity does

Uncorrelated but dependent is possible ($Y = X^2$)

Larger PDF peak ⇒ smaller σ; P(A=0) = 0 for continuous A

n = 30 is not automatically enough for the CLT

Calibration drift breaks the LLN conclusion

Using s instead of σ ⇒ t-distribution

Halving the margin of error ⇒ quadruple n